

HEALTHCONNECT CLINIC

Analysis of factors associated with missed appointments

Programme : AnalystLab Africa — Experience Lab

Course: Data Analytics

Projet : HealthConnect Clinic

Period: Week 5

Dataset: [HealthConnect_Appointment_Data.csv](#) **Main tools:** Power BI Desktop, Power Query, DAX, Python **Role:** Junior Data Analyst

1. CONTEXT AND PROBLEM STATEMENT

HealthConnect Clinic seeks to reduce the number of missed appointments in order to improve the use of its available slots, better understand patient attendance behaviors, and enhance the efficiency of administrative follow-up.

The analysis is based on a fictitious and anonymized dataset containing **5,000 medical appointments**.

The main analytical question is as follows:

What observable factors in the data are associated with a higher probability of missed appointments, and how can this information help HealthConnect better target its follow-up actions?

The objective of this analysis is not to demonstrate a causal relationship between variables and absences, but to identify **the most important associations** in order to generate courses of action for the following stages of the project.

2. ANALYTICAL OBJECTIVES FOR WEEK 5

The work of Week 5 aims to transform the hypotheses identified during Week 4 into concrete analytical results.

The objectives selected are:

1. Check and prepare the quality of the dataset.
2. Finalize the transformations necessary for the analysis.
3. Construct the main performance indicators (KPIs).
4. Explore the relationships between the characteristics of the appointments and their outcome.

5. Identify the variables showing the strongest associations with No-Shows.
6. Build an initial analytical dashboard in Power BI.
7. Formulate initial business interpretations.
8. Identify the limitations of the analysis and areas for further exploration for Week 6.

3. DATASET DESCRIPTION

The [HealthConnect_Appointment_Data.csv](#) file contains:

- **5 000 observations**

- **18 variables** • an

observation corresponding to **a medical appointment** • `appointment_id` as a unique identifier of the appointment • `patient_id` allowing to link several appointments to the same patient.

The variables cover several dimensions:

Patient information

- `patient_id` •
`gender`
- `age`
- `age_group`

Appointment information

- `appointment_id` •
`appointment_type` •
`appointment_date` •
`appointment_day` •
`appointment_time` •
`appointment_outcome`

Booking information

- `booking_date` •
`booking_lead_days`

Patient history

- `previous_appointments` •
`previous_no_shows`

Reminders

- `reminder_sent`

- reminder_channel

Accessibility and experience

- distance_to_clinic_km
- waiting_time_minutes

4. INITIAL DATA QUALITY AUDIT

The audit was conducted using **Power Query**, specifically through column profiling and quality control features.

4.1 General Structure

The dataset contains the expected 5,000 appointments and no rows were deleted during the cleanup.

No duplicates were identified on:

- appointment_id • the complete rows of the dataset.

The appointment_id key therefore contains 5,000 unique values.

4.2 Missing values

Three variables have zero values:

Variable	Values missing	Rate
reminder_channel	1 366	27,32 %
distance_to_clinic c_km	90	1,80 %
waiting_time_minu hand	60	1,20 %

reminder_channel

The 1,366 null values do not correspond to a random absence of data.

The verification shows that they correspond to the appointments for which:

reminder_sent = "No"

Therefore, it is more relevant to interpret these values as:

No recall

rather than as a missing piece of information.

This distinction is important because removing these lines would have resulted in a loss of information on the group of patients who did not receive any reminders.

distance_to_clinic_km

90 values are missing, representing 1.80% of the dataset.

The non-response mechanism has not been established with sufficient evidence to formally conclude that it is an MCAR mechanism.

Given the small volume involved and the need to retain the observations, imputation by the **median** was chosen.

Observed median: **8.7 km**

The zero values were therefore replaced by **8.7 km**.

This decision limits the influence of extreme values compared to imputation by the mean.

waiting_time_minutes

60 values are missing, representing 1.20% of the dataset.

As with distance, the non-response mechanism has not been statistically demonstrated.

The median observed time is: **24 minutes**

The zero values were therefore replaced by **24 minutes**.

5. CONSISTENCY CHECKS

Several checks were carried out before the analysis.

Dates

The booking and appointment dates are consistent:

no appointment has a booking date later than its appointment date.

The variable `booking_lead_days` is also consistent with the difference between `booking_date` and `appointment_date`.

Historical

The number of previous_no_shows never exceeds the number of previous_appointments.

Age

The patients' ages range from **18 to 80 years**, in accordance with the adult scope of the dataset.

Waiting time

The available values for waiting_time_minutes are positive.

6. Data Transformation in Power Query

6.1 Date Conversion

The variables:

- booking_date •
- appointment_date

were initially interpreted as textual values.

Since the file format was American MM/DD/YYYY, a direct conversion with the French locale could cause a misinterpretation of the dates.

The transformation used in Power Query is:

Change type ÿ Use locale settings ÿ Type: Date ÿ Locale: English (United States)

This transformation allows dates to be correctly converted into true Date type values.

7. ADDRESSING MISSING VALUES

7.1 Reminders

The null values of reminder_channel have been replaced by:

None

This transformation allows us to maintain the distinction between:

- no reminder;

- Email reminder; •
 - SMS reminder; •
 - WhatsApp reminder.
-

7.2 Distance

The 90 zero values were replaced by the median : **8.7 km**

The median was preferred because it is less sensitive to extreme values.

7.3 Waiting Time

The 60 zero values have been replaced by : **24 minutes**

The same robust imputation logic was applied.

8. FEATURE ENGINEERING

To facilitate analysis in Power BI, two categorical variables were created.

8.1 Booking deadline

Variable source :

booking_lead_days

Segmentation :

Condition	Category
0–7 days	1. 0–7 days
8–14 days	2. 8–14 days
15–30 days	3. 15–30 days
>30 days	4. More than 30 days

This transformation makes it easier to study the relationship between booking lead time and no-show rate.

8.2 Distance range

Variable source :

distance_to_clinic_km

Segmentation :

Condition	Category
≤5 km	1. Less than 5 km
>5 et ≤10 km	2. 5–10 km
>10 et ≤20 km	3. 10–20 km
>20 km	4. More than 20 km

Numeric prefixes help maintain the logical order of categories in Power BI.

9. DEFINITION OF THE TARGET VARIABLE

The `appointment_outcome` variable contains three modalities:

- Attended
- No-Show
- Cancelled

For the main analysis, the **No-Show** constitutes the event of interest.

It is important to distinguish between cancelled appointments and missed appointments:

A cancellation is not considered a no-show.

The project's main KPI is nevertheless retained according to the definition used since Week 4:

No-Show Name / Full Rendezvous Name

This definition guarantees the methodological continuity of the project.

10. THE KPI OF THE POWER MODE

10.1 Total appointment volume

Total Appointments =

COUNTROWS('HealthConnect_Appointment_Data')

Result: 5,000

10.2 No-Show Name

No-Show Name =

```
CALCULATE(  
    COUNTROWS('HealthConnect_Appointment_Data'),  
    'HealthConnect_Appointment_Data'[appointment_outcome] = "No-Show"  
)
```

Result: 2,423

10.3 Overall No-Show Rate

No-Show Rate =

```
DIVIDE(  
    [Name No-Show],  
    [Total Appointments], 0  
)
```

Result: 48.46%

This measure is formatted as a percentage in Power BI.

11. INITIAL RESULTS OF THE ANALYSIS

11.1 Overall Rate

Out of 5,000 appointments:

- **2,423** are no-shows; • **2,314** are honored appointments; • **263** are cancelled.

The no-show rate reached:

48,46 %

This level justifies a thorough analysis of the factors associated with absences.

12. ANALYSIS OF THE BOOKING DEADLINE

The time between booking and the date of the appointment shows a particularly clear relationship with the no-show rate.

Deadline	Rate of No-Show
0–7 days	27,81 %
8–14 days	33,55%
15–30 days	43,21%
>30 days	60,49 %

A steady progression is observed.

The no-show rate thus increases from **27.81%** for appointments booked within 7 days to **60.49%** when the delay exceeds 30 days.

Interpretation

The booking lead time appears to be a variable strongly associated with the risk of no-show.

However :

This observation does not allow us to conclude that the delay directly causes the absence.

Other characteristics of patients or appointments may contribute to this relationship.

13. ANALYSIS OF NO-SHOW'S HISTORY

Previous No-Shows	Rate of No-Show
0	43,51 %
1	53,49 %
2	59,36 %
3+	68,82 %

A gradual increase in the No-Show rate is observed as the number of previous absences increases.

Patients with at least three previous No-Shows have a No-Show rate of **68.82 %**.

Interpretation

Therefore, **previous_no_shows** constitutes a particularly interesting behavioral signal for identifying groups at higher risk.

This variable will need to be further investigated in future analyses, particularly in combination with the booking lead time.

14. DISTANCE ANALYSIS

Distance	Rate of No-Show
≤5 km	46,45 %
5–10 km	46,51 %
10–20 km	49,43 %
>20 km	57,76 %

The relationship appears more moderate than that observed for delay and history.

The rate remains relatively close up to 20 km, then increases to **57.76%** beyond 20 km.

Interpretation

Distance could be an additional factor, particularly for patients located more than 20 km away.

A cross-analysis with other variables will be necessary before making it a priority operational lever.

15. ANALYSIS OF RECALLS

Situation	Rate of No-Show
No recall	51.39%
Reminder sent	47,36 %

The group that received a recall has a No-Show rate that is approximately **4 points lower**.

By channel:

Canal	Rate of No-Show
SMS	45,75 %
Email	48,41 %
WhatsApp	49,77 %

Interpretation

The data shows an association between sending a reminder and a lower no-show rate.

However, this observation does not allow us to conclude that the recalls are directly responsible for this decrease.

Selection bias may exist in particular: patients receiving a reminder may differ from patients not receiving one on other characteristics.

The analysis of the recall will therefore need to be further refined by cross-referencing with:

- booking lead time; • no-show history; • distance; • appointment type; • possibly day and time.

16. HIERARCHY OF VARIABLES

Based on the exploratory analysis, the variables can be grouped into three levels.

High priority

1. **booking_lead_days**

Particularly strong relationship with the no-show rate.

2. **previous_no_shows**

Significant behavioral signal and steady increase in the absence rate.

Intermediate priority

3. **distance_to_clinic_km**

The effect is more moderate, but particularly visible beyond 20 km.

4. reminder_sent / reminder_channel

An interesting association, but one that requires controlled analysis before any operational conclusions are drawn.

Exploratory priority

Variables such as:

- age; •
type of appointment; • day;
• time; •
waiting
time;

individually present less significant discrepancies in the initial analysis.

However, they should not be definitively eliminated: their interaction with priority variables can reveal specific profiles.

17. KEY BUSINESS QUESTIONS

The analysis helps to structure the next steps around five questions:

1. What is HealthConnect's overall No-Show rate?
2. How is the booking lead time associated with the no-show rate?
3. Does the No-Show history allow for the identification of patients at high risk?
4. Does distance from the clinic contribute to the risk of absence?
5. Are reminders associated with a decrease in the no-show rate, and is there a difference depending on The canal?

18. POWER BI DASHBOARD

The analytical dashboard was built to allow for a quick overview of the situation and an exploration of the main factors associated with absences.

The main elements include:

- KPI of total appointment volume; • KPI of
number of no-shows; • KPI of no-
show rate; • visualization of no-
show rate according to booking time; • visualization according to no-show
history; • distance analysis; • reminder analysis; •
filters allowing exploration of
different segments.

The dashboard provides an initial layer of analytical reporting. It will be enhanced if the cross-analyses from Week 6 reveal new needs.

19. VALIDATION FINALE — QA CHECK

After transformation:

- **5,000 rows retained** • no unjustified deletions • **0 duplicates identified** • no residual null values in the processed variables
- Dates correctly converted
- Consistency between dates and **booking_lead_days** • Consistency between **previous_no_shows** and **previous_appointments**

The reference figures are:

KPI	Value
Total appointments: 5,000	
No-Show	2 423
Attended	2 314
Cancelled	263
No-Show Rate	48,46 %

These values constitute the references used in the dashboard.

20. LIMITATIONS OF THE ANALYSIS

Several limitations must be taken into account.

20.1 Association and Causality

The analyses performed are primarily descriptive and bivariate.

A difference in rates between two groups does not necessarily mean that one variable is the cause of the No-Show.

20.2 Fictitious data

The dataset is fictitious and anonymized. Therefore, the results cannot be directly generalized to a real patient population.

20.3 Missing values

The non-response mechanisms of `distance_to_clinic_km` and `waiting_time_minutes` have not been formally established.

The imputations performed allow the observations to be retained, but may introduce a slight modification to the distribution of these variables.

20.4 Potentially confounding variables

Some observed relationships may be influenced by other variables.

For example, the relationship between callback and no-show may depend on the booking lead time or the patient's behavioral profile.

Multivariate analyses would be necessary to better isolate the effects.

21. IMPORTANT ANALYTICAL DECISIONS

Several decisions made during the analysis were retained in order to ensure the traceability of the project.

Decision 1 — Retention of cancelled appointments

`Cancelled` appointments have not been removed from the source dataset.

They remain present in the data in order to preserve the integrity of the dataset.

Decision 2 — Preservation of values without recall

Null values of `reminder_channel` have been transformed into category `None`, because they correspond to `reminder_sent = No`.

Decision 3 — Imputation by median

Missing values for distance and waiting time were replaced by their respective medians to limit the influence of extreme values.

Decision 4 — Use of categories for continuous variables

The booking time and distance have been transformed into slices to improve their interpretation in visualizations and their use by business users.

22. KEY INSIGHTS TO REMEMBER

The analysis of Week 5 highlights four main lessons:

Insight 1

The overall no-show rate is high : **48.46% of appointments**

Insight 2

The booking lead time is strongly associated with No-Show : **27.81%** at 0–7 days versus **60.49%** beyond 30 days.

Insight 3

Behavioral history is also strongly associated with No-Show : **43.51%** without previous No-Show versus **68.82%** for patients with 3 or more previous No-Shows.

Insight 4

Distance shows a more moderate association but becomes notable beyond 20 km : **57.76%** No-Show.

Insight 5

Appointments that received a reminder had a lower no-show rate : **47.36% compared to 51.39%**.

but this difference must be interpreted with caution because it does not demonstrate a causal effect of the recall.

23. GUIDELINES FOR THE NEXT STEPS

The results of this step allow us to move from a descriptive analysis to a more targeted analysis.

The next steps should focus in particular on:

1. Cross-analysis of **booking lead time and no-show history** ;
2. Identification of profiles presenting multiple risk factors simultaneously;
3. More rigorous analysis of the apparent effect of reminders;
4. Study of potential interactions with the day, time, and type of appointment;
5. Definition of operational patient segments;
6. Improvement of the dashboard based on analytical results;
7. Formulation of business recommendations based on the results.

The goal is therefore not simply to note that certain variables are associated with No-Shows, but to determine **how HealthConnect could use this information to prioritize its actions.**

24. CONCLUSION

Week 5 enabled the transformation of the hypotheses identified during Week 4 into a first structured and visualized analysis.

The cleaning process resulted in a usable dataset while retaining the initial 5,000 observations.

The exploratory analysis particularly highlights two variables:

- the **booking lead time** ; •
- the No-Show history.**

Distance and reminders also constitute interesting areas of analysis, but require more cautious interpretation.

The Power BI dashboard now provides an initial analytical view of the HealthConnect situation.

The results obtained form a basis for the next phase of the project: to deepen the interactions between variables, identify risk profiles and transform the analytical results into **operational recommendations for HealthConnect Clinic.**